

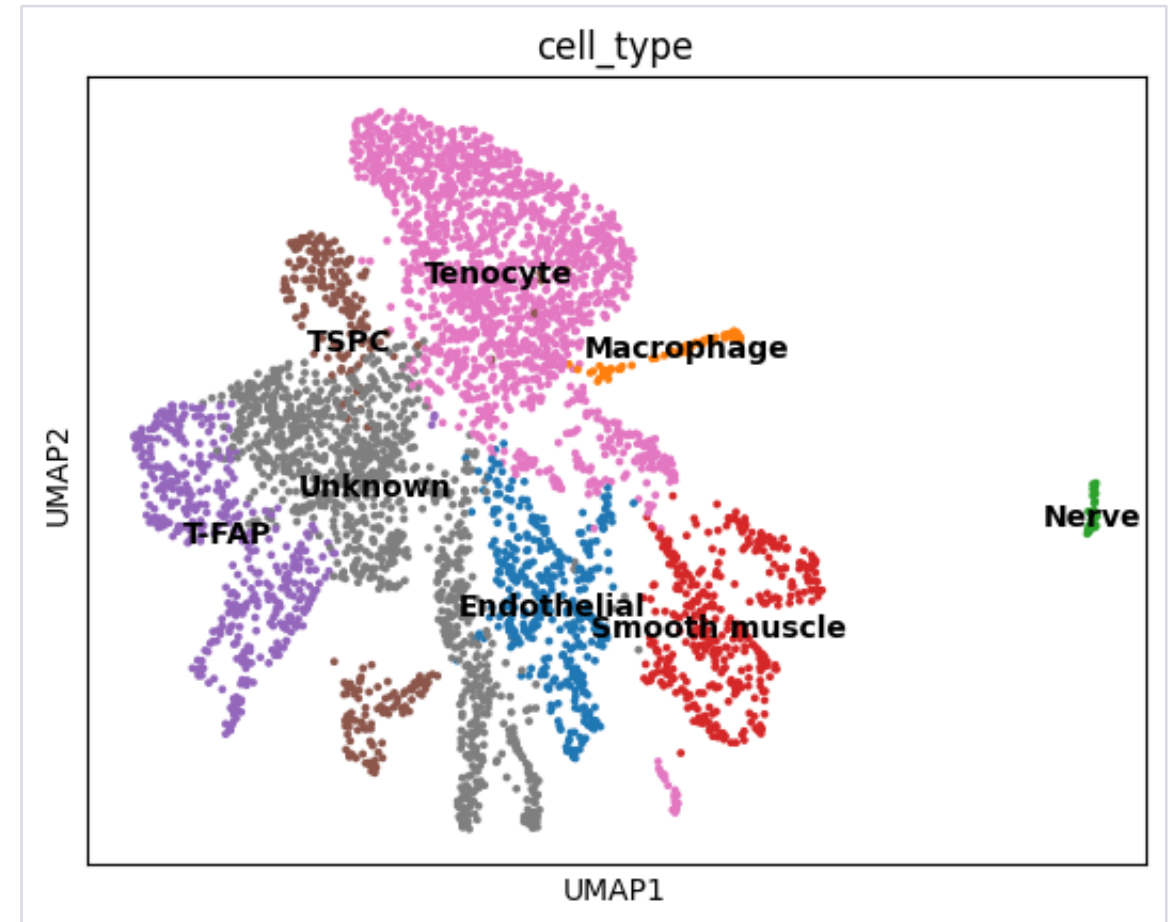
A Mechanism-Informed Gene Signature for Regenerative vs. Fibrotic Tendon Healing

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NARRATIVE DRAFT — figures/tables preliminary, per storyboard rubric

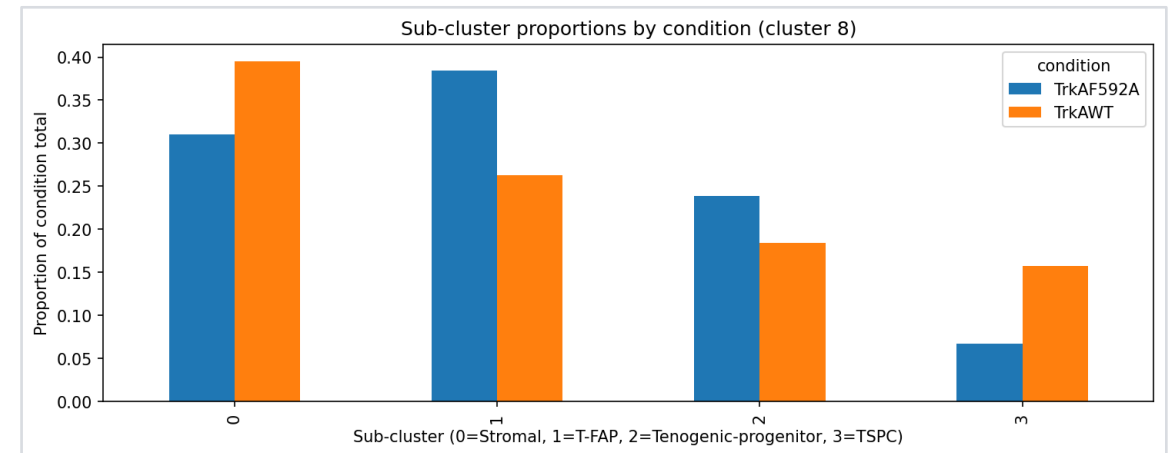
Tendons heal by one of two divergent routes — regeneration or fibrosis — and the cellular basis was only recently resolved.

- Tendon injuries are common and often heal poorly, and the tendon sheath is now known to contain two co-resident PDGFR α + progenitor populations that determine which way healing goes.
- Tppp3+ Tendon Stem/Progenitor Cells (TSPCs) restore normal fibrillar tendon architecture, while Tppp3- Tendon Fibro/Adipogenic Progenitors (T-FAPs) instead produce permanent fibrotic scar tissue.
- This TSPC/T-FAP dichotomy was established by Harvey et al. (2019) using single-cell RNA sequencing of uninjured mouse tendon — the field's first cellular explanation for why some tendons regenerate and others scar.



Innervation status controls which fate progenitors take, but the downstream molecular program executing that choice was uncharacterized.

- Cherief et al. (2023) showed that surgically denervating an injured mouse tendon blocks TSPC expansion and shifts healing toward the fibrotic T-FAP outcome, implicating sensory nerve signalling (TrkA/NGF) as an upstream control switch.
- This establishes that innervation controls the fate decision, but not how: the downstream transcription-factor programs that actually execute the TSPC-vs-T-FAP choice inside the cell were never mapped.
- Closing that mechanistic gap is the first objective of this project.



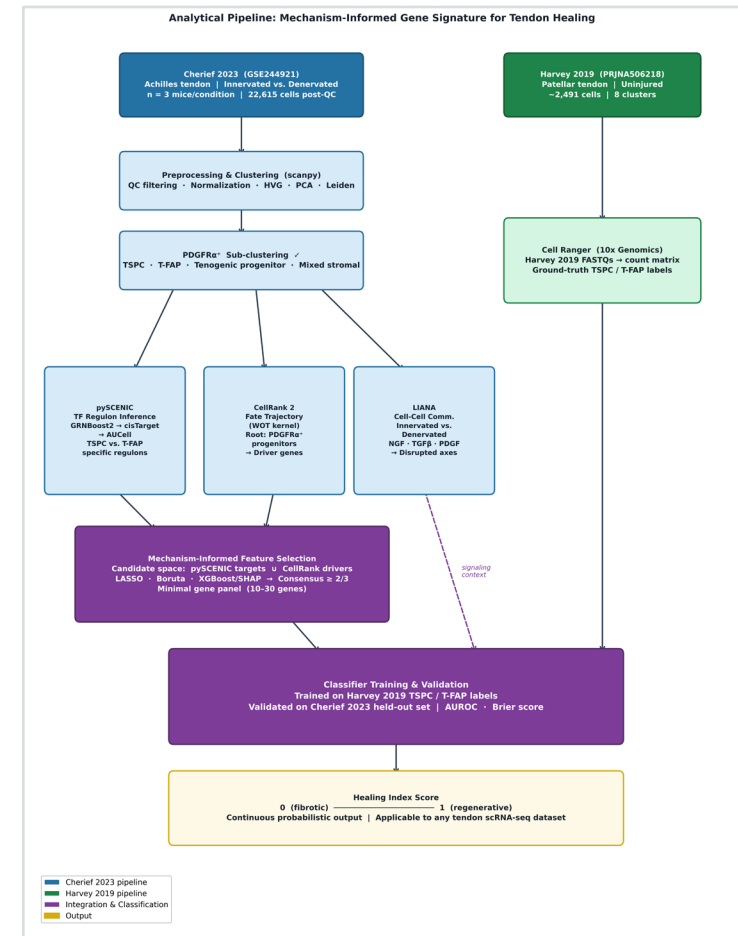
TSPC/T-FAP proportions shift with innervation status

A second gap: no validated computational tool exists to objectively score a tendon sample's healing trajectory.

- Even once a tendon has healed, there is currently no objective way to say how well it healed — clinical and experimental assessment still relies on subjective histological scoring by a human observer.
- TGF β (Kaji et al. 2020) and PDGF signalling have also been implicated as fate-control signals, reinforcing that this is a multi-signal decision, not a single-pathway switch.
- What's missing is a reusable, quantitative readout — a score computable directly from a tendon's gene expression profile.

This project closes both gaps: map the TSPC/T-FAP regulatory circuit, then compress it into a validated, continuous Healing Index.

- Hypothesis: innervated-niche signals (NGF, TGF β , PDGF) activate a set of TSPC-biasing transcription-factor regulons; when denervation removes these signals, T-FAP-driven fibrosis proceeds by default.
- Objective 1 (mechanistic arm): map the TSPC/T-FAP regulatory circuit using pySCENIC, CellRank, and LIANA.
- Objective 2 (applied arm): compress the resulting mechanism-derived gene programs into a minimal, validated classifier that outputs a continuous Healing Index.
- Novelty: the first TF-level regulatory circuit connecting innervation status to tendon progenitor fate, and the first quantitative healing score proposed for this system.



Full two-arm analytical pipeline

Two published mouse tendon scRNA-seq datasets anchor the analysis, chosen because each supplies something the other cannot.

Dataset	Source	Cells used	Role in project
Cherief 2023	GEO GSE244921	22,615 (cluster 8: 3,900 sub-clustered)	Mechanistic analysis (pySCENIC, CellRank, LIANA)
Harvey 2019	SRA PRJNA506218	4,069 (TSPC 266 / T-FAP 433)	Classifier training + ground-truth labels

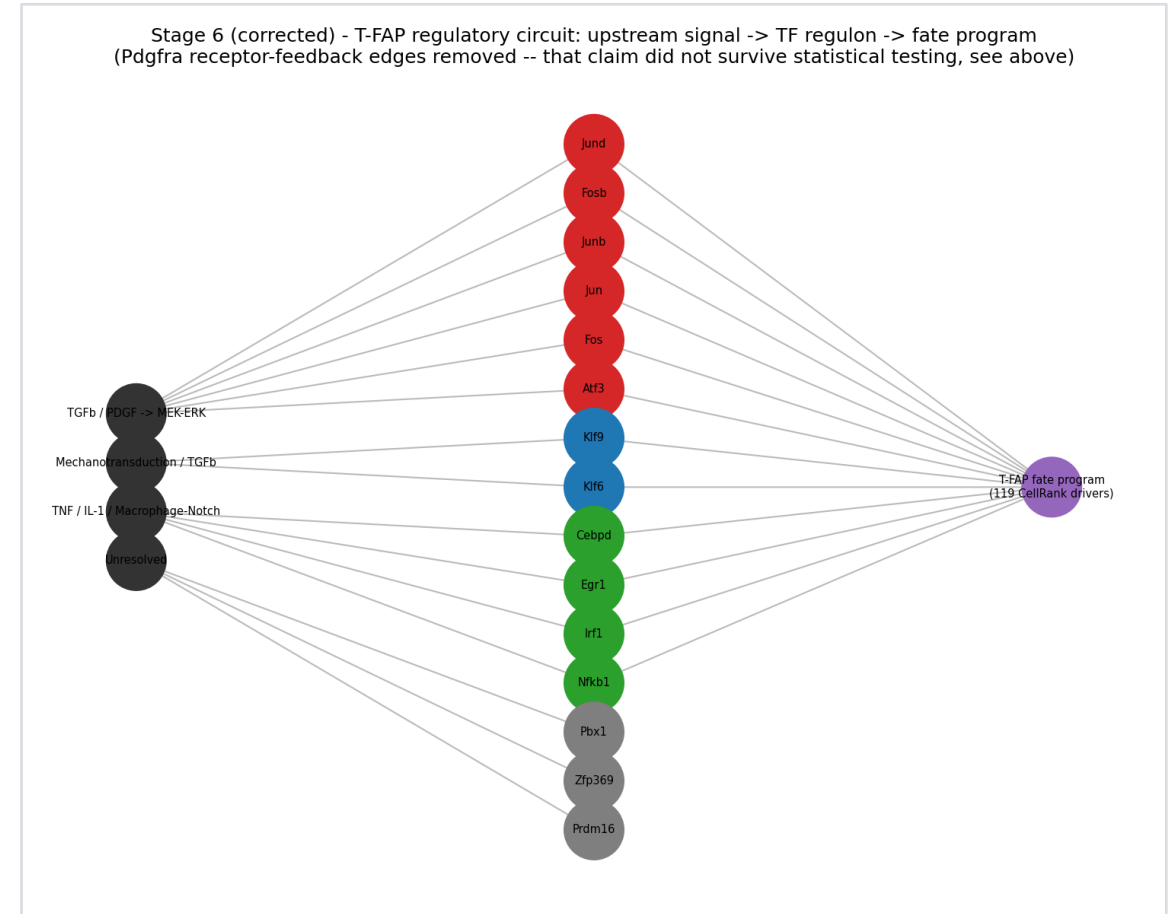
Mechanism and classifier training draw on two independent datasets: Cherief 2023 supplies the innervated-vs-denervated contrast needed for mechanism; Harvey 2019 supplies independently validated TSPC/T-FAP labels needed as classifier ground truth.

Python-based single-cell and machine-learning stack, chosen so each tool plays a distinct, non-overlapping role.

- scanpy handles quality control, normalization, and clustering for both datasets, following the standard single-cell preprocessing workflow.
- pySCENIC infers transcription-factor regulatory networks via a three-step pipeline: GRNBoost2 co-expression modules, cisTarget motif-enrichment pruning, and AUCell per-cell regulon scoring.
- CellRank 2 models fate trajectories using a pseudotime kernel and GPCCA macrostate analysis, producing per-cell fate probabilities and lineage-driver genes.
- LIANA infers cell-cell communication by aggregating six independent methods into one consensus ranking.
- scikit-learn, Boruta, and XGBoost together perform feature selection, chosen because they represent three different modelling assumptions (penalized linear, tree-ensemble wrapper, gradient boosting).

Three independent mechanistic analyses, cross-referenced into one regulatory circuit.

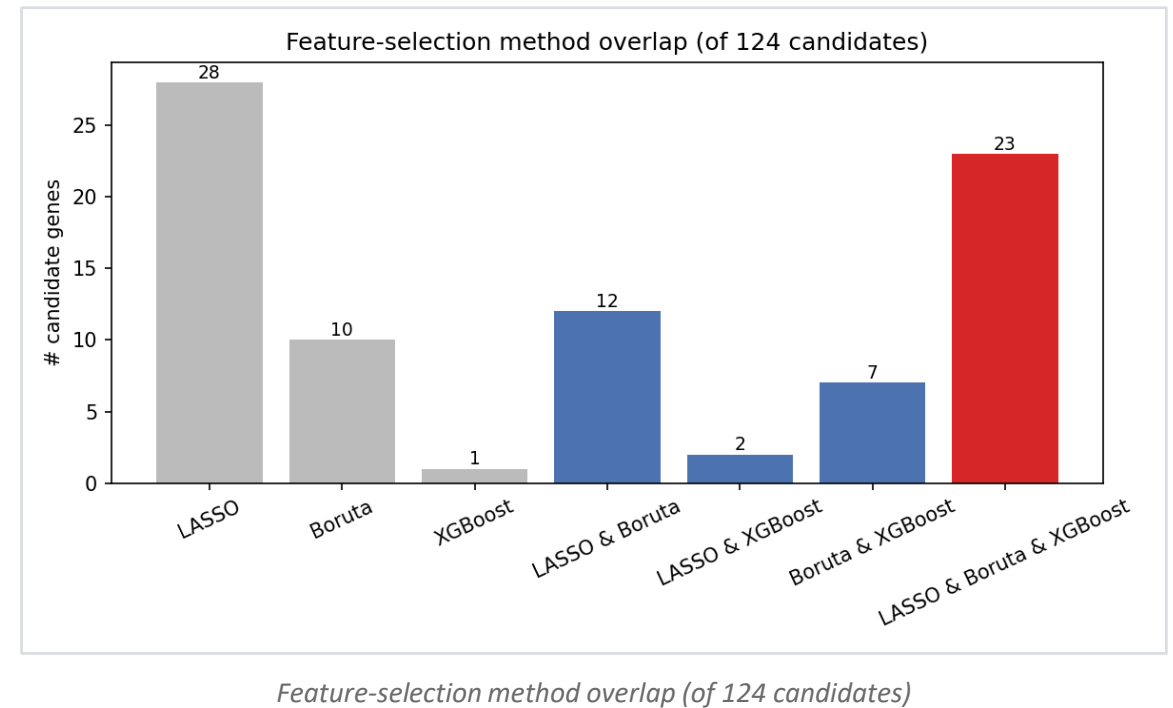
- pySCENIC (Van de Sande et al. 2020) infers TF regulons independently in each dataset; a Wilcoxon test with a cross-dataset direction filter then identifies regulons consistently associated with TSPC or T-FAP in both Harvey and Cherief.
- CellRank 2 (Weiler et al. 2024) combines a diffusion-pseudotime kernel with GPCCA macrostate decomposition on the Cherief cluster-8 population, estimating each cell's probability of committing to a TSPC or T-FAP fate.
- LIANA (Dimitrov et al. 2022) compares ligand-receptor signalling between innervated and denervated conditions using a six-method consensus ranking, focused on TSPC/T-FAP as receiving cell types.
- All three outputs are cross-referenced by shared target genes into a single circuit linking upstream signal family, TF regulon, and the fate program those regulons drive.



Signal family → TF regulon → T-FAP fate program

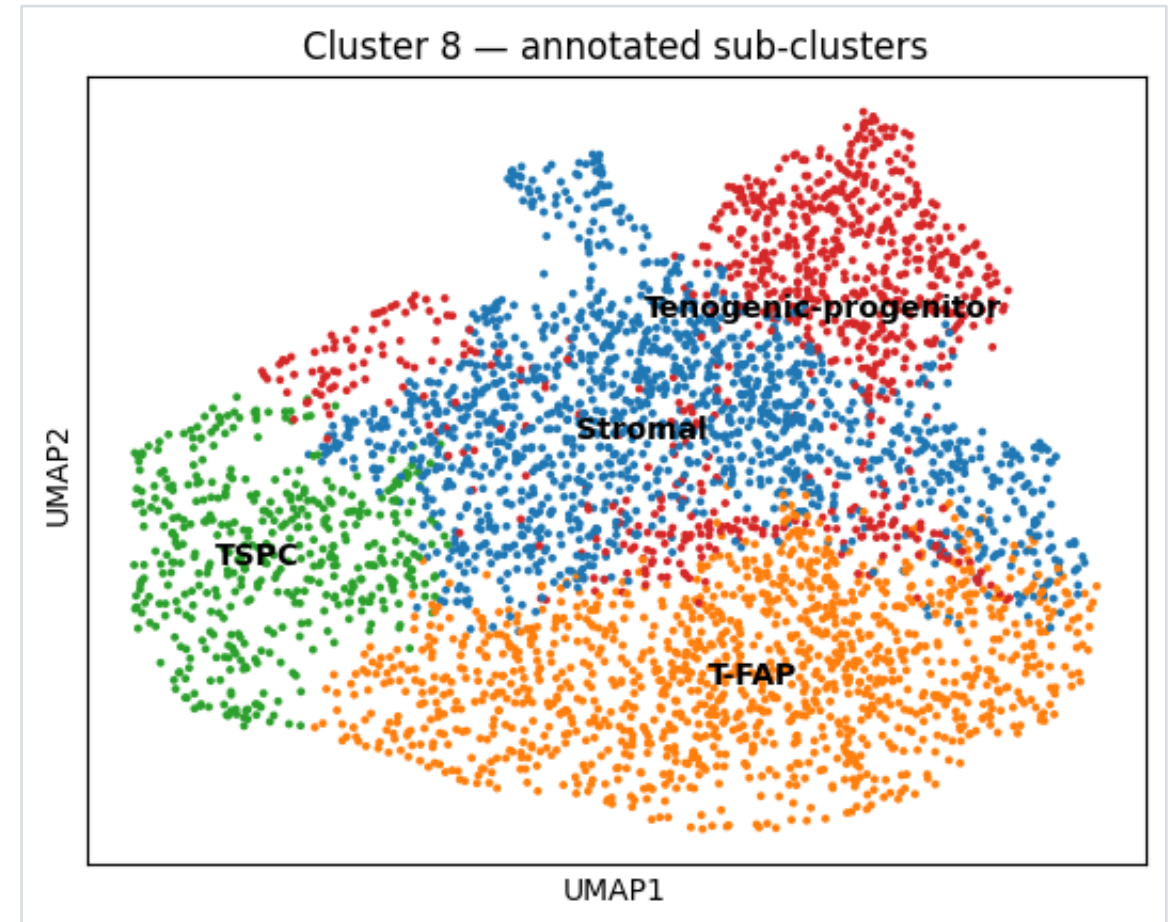
Feature selection and classifier: three ML methods narrow 124 mechanism-derived candidates to a minimal panel.

- The candidate gene pool is deliberately narrow and mechanism-derived rather than genome-wide: the 119-gene overlap between pySCENIC T-FAP regulon targets and CellRank T-FAP lineage-driver genes, plus 5 established TSPC markers — 124 candidates total.
- LASSO (10-fold cross-validated), Boruta (a random-forest wrapper tested against randomized shadow features), and XGBoost (gradient-boosted trees, thresholded by gain importance) each select candidates completely independently.
- Genes chosen by at least 2 of the 3 methods form a 44-gene working panel; genes chosen by all 3 form a stricter, unanimous 23-gene core panel.
- A logistic regression classifier is trained on Harvey and validated on the independent Cherief dataset; its calibrated output probability becomes the continuous Healing Index.



QC and clustering recover the expected tendon cell landscape, with one PDGFR α + population containing both progenitor states.

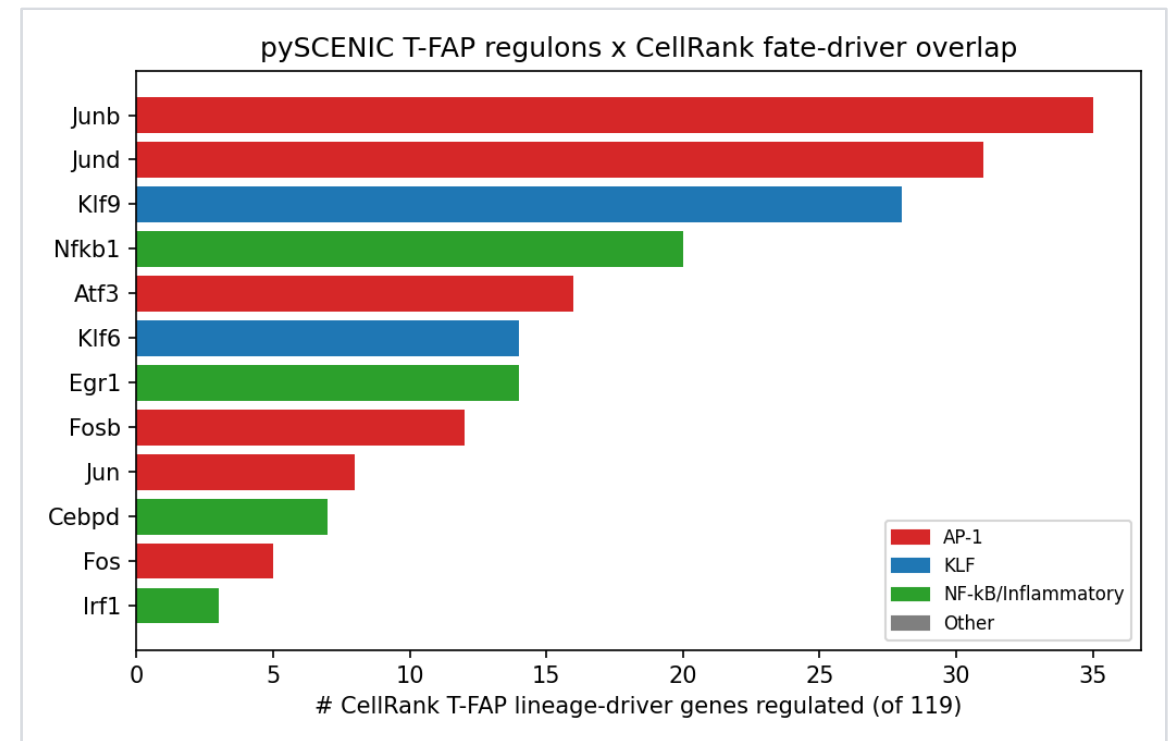
- Standard QC filtering retained 22,615 of 25,545 Cherief cells (88.5%), grouped into 17 Leiden clusters.
- One cluster — cluster 8, marked by PDGFR α expression, containing 3,900 cells — is the focus of this project because it is the only cluster where TSPC and T-FAP are mixed together rather than already separated.
- Sub-clustering cluster 8 at higher resolution cleanly separated it into four populations: TSPC (451), T-FAP (1,245), Tenogenic-progenitor (816), and Stromal (1,388).



Cluster-8 sub-populations, annotated

TF network inference finds a robust 15-regulon fibrotic program and zero consistent TSPC regulons — a result that survived an independent robustness audit.

- Independent inference in each dataset produced 471 Harvey regulons and 417 Cherief regulons; testing which were consistently associated with T-FAP across both left 15 — but zero TSPC regulons replicated.
- The 15 T-FAP regulons cluster into three families — AP-1, KLF, NF-κB/inflammatory — and Junb, Jund, and Klf9 alone account for most CellRank-defined T-FAP driver genes.
- Because TSPC has far fewer cells than T-FAP in both datasets, this could plausibly be a statistical power artifact, so it was tested directly with a 200-iteration power-matched audit, which reproduced 14/15 T-FAP regulons and confirmed the TSPC result is not a sample-size artifact.



CellRank T-FAP driver genes per TF regulon

Three independent feature-selection methods converge on a 23-gene core panel that recovers all five hand-picked TSPC markers.

- Of 124 mechanism-derived candidates, LASSO selected 65, Boruta confirmed 52, and XGBoost selected 33, each working independently; requiring agreement from all 3 narrows this to the 23-gene core panel below, which recovers all 5 hand-picked TSPC markers and is 68% Junb/Jund/Klf9 targets.

Gene	Notable role
C3	Junb/Jund/Klf9 target
Camk2n1	Junb/Jund/Klf9 target
Cpxm1	Junb/Jund/Klf9 target
Fth1	Junb/Jund/Klf9 target
Igfbp7	Junb/Jund/Klf9 target
Nfib	Junb/Jund/Klf9 target
Ptx3	Junb/Jund/Klf9 target
Rps29	Junb/Jund/Klf9 target
Scn7a	Junb/Jund/Klf9 target
Serping1	Junb/Jund/Klf9 target
Slc38a2	Junb/Jund/Klf9 target
Slfn5	Junb/Jund/Klf9 target

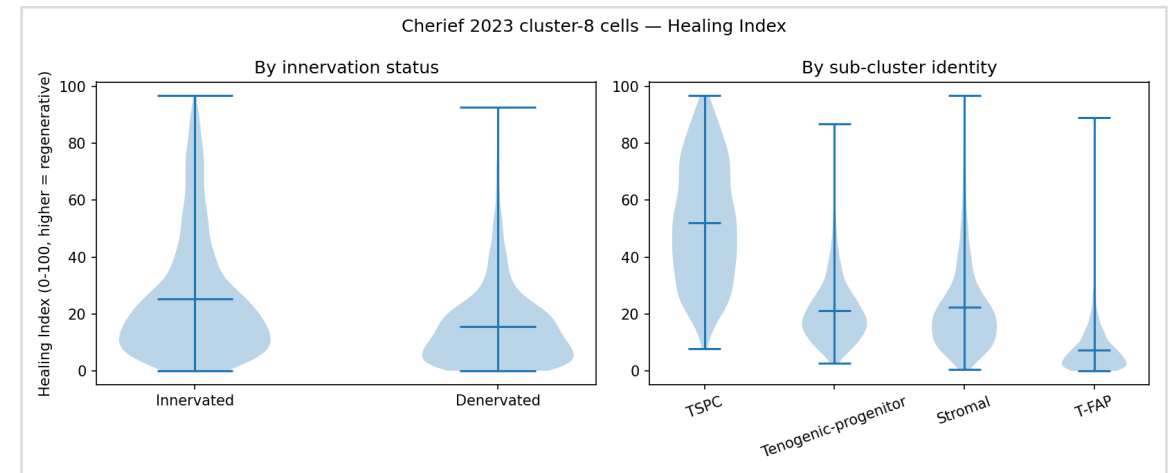
Gene	Notable role
Steap4	Junb/Jund/Klf9 target
Zfp36l1	Junb/Jund/Klf9 target
Aebp1	—
C1ra	—
Cebpd	—
Cxcl12	—
Lpl	—
Sema3c	TSPC marker
Sfrp2	—
Tppp3	TSPC marker
Zfp36	—

Deliverable — summary table: 23-gene core panel (unanimous LASSO + Boruta + XGBoost agreement).

Paired visualization deliverable: method-overlap chart (previous slide) and Healing Index chart (next slide).

The classifier generalizes to an independent dataset and produces a continuous Healing Index that tracks innervation status.

- Trained on Harvey and evaluated by 10-fold cross-validation, the core panel reaches ROC-AUC 0.993 — but that alone doesn't prove generalization, since it never leaves the training dataset.
- The real test: applying the Harvey-trained classifier, unchanged, to Cherief's independently labelled TSPC/T-FAP cells reaches ROC-AUC 0.989, accuracy 93.5%, on genuinely held-out data.
- The resulting Healing Index (0–100, higher = more regenerative) shows the expected pattern: mean 25.2 in innervated tissue vs. 15.5 in denervated tissue.
- This Healing Index also correlates with CellRank's independent fate-probability estimate (Spearman $\rho = 0.656$) — two unrelated methods partially agreeing on the same biology.



Healing Index by condition and sub-cluster

The project delivers the three outcomes proposed, with the classifier exceeding its pre-registered success bar.

- The original proposal set a pre-registered success threshold of held-out AUROC > 0.85; the actual cross-dataset result of 0.989 clears that bar by a wide margin.
- All three promised deliverables were produced: a mechanistic TF-to-fate regulatory circuit, a validated 23-gene minimal gene panel, and a continuous Healing Index applicable in principle to any new tendon scRNA-seq dataset.
- This matters because it shows that mechanism-derived features — genes chosen because they sit inside a biologically motivated circuit, not because they scored well in a black-box search — can still produce a strong, interpretable classifier.

Rigor mattered as much as discovery — three headline findings were explicitly stress-tested, and not all survived.

- Three headline findings from earlier stages were deliberately re-tested with more rigorous statistics before being trusted — the outcomes were mixed, which is the point.
- Reporting retractions alongside confirmations, not just what worked, is what makes the surviving findings — including the gene panel and classifier — more trustworthy.

Finding	Test applied	Outcome
"Efna1→Epha3 strongest signal"	Permutation test (n=100)	Retracted (p=1.0)
Pdgfra receptor feedback	Hypergeometric + BH correction	Retracted (n.s.)
Zero consistent TSPC regulons	200-iter power-matched audit	Confirmed (14/15 replicate)

Discussion visual: every headline finding stress-tested, not all survived.

Findings extend prior literature by identifying the specific TF program executing the innervation-dependent fate decision, plus a reusable quantitative score.

- Cherief et al. (2023) established that innervation controls the TSPC/T-FAP fate decision; this project's contribution is identifying which transcription-factor program executes that decision — an AP-1/KLF network anchored by Junb, Jund, and Klf9.
- This is a concrete, testable mechanistic hypothesis where previously there was only a phenomenological observation, consistent with the broader TGFβ/PDGF signalling context established by Kaji et al. (2020) and Harvey et al. (2019).
- The translational motivation for the Healing Index is to move tendon-healing assessment away from subjective histological scoring and toward an objective, reproducible, gene-expression-based score.
- A turnkey scoring function is already built and ready to apply to genuinely new lab-generated tendon scRNA-seq data as soon as it becomes available.

Key limitations temper interpretation.

- Neither Harvey 2019 nor Cherief 2023 includes biological replicates — each is a single pooled sample per condition — so every validation performed (cross-validation within Harvey, generalisation to Cherief, the CellRank cross-check) is a cross-dataset comparison, not a test across independent animals or cohorts.
- Harvey's own T-FAP cluster was excluded from trajectory analysis by the original paper's authors, a caveat that carries forward into every downstream analysis relying on that label.
- The classifier is asymmetric in a way worth stating plainly: TSPC recall is only 77–78% vs. 98–99% for T-FAP, meaning the Healing Index is more reliable at flagging fibrotic drift than at confidently confirming full regenerative identity.
- The next, and arguably most important, step is applying this pipeline to genuinely new lab-generated tendon scRNA-seq — the first real test across independent biological samples this signature will face.

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